

NITHWIN V M

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SUMMARY

Computer Science graduate with experience in backend engineering, Generative AI, and developer tools. Built WindMist, an open-source AI coding assistant in Go, scalable AI applications, and computer vision systems. Solved 770+ LeetCode problems and won 2× First Prize in university innovation competitions.

EDUCATION

Nandha Engineering College

Bachelor of Engineering in Computer Science and Engineering
CGPA: 8.2/10

Erode, TN

2022 – 2026

TECHNICAL SKILLS

- **Languages:** Go, C++, Java, Python, TypeScript, JavaScript, SQL
- **Backend:** Node.js, Express.js, RESTful APIs, WebSockets
- **Frontend:** Next.js, React.js, Tailwind CSS
- **Databases:** PostgreSQL, SQLite, Prisma ORM, ChromaDB
- **AI & Generative AI:** RAG, LangChain, MCP, AI Agents, Tool Calling, Gemini API, Groq API, FAISS
- **Developer Tools:** Git, GitHub, Linux, Docker, GitHub Actions, CI/CD, Postman, Vercel

PROJECTS

WindMist — Open-Source AI Coding Assistant | github.com/Nithwin/windmist | windmist.vercel.app

- Built WindMist, an open-source AI coding assistant for the terminal in Go with a modular provider architecture supporting multiple LLM providers.
- Developed repository-aware file editing tools, streaming AI responses, Markdown rendering, and intelligent context retrieval for code understanding.
- Implemented provider registry, configuration management, automated testing, GitHub Actions CI/CD, and automated cross-platform releases.
- **Tech Stack:** Go, Cobra CLI, Bubble Tea, Lip Gloss, Glamour, Docker, GitHub Actions, Gemini API

ShadowCoders - Scalable Examination Platform | github.com/Nithwin/ShadowCoders

- Built a full-stack online examination platform serving 400+ students, supporting reliable real-time examinations for 80+ concurrent users.
- Reduced backend workload by approximately 90% through secure client-side code execution and implemented 11 browser-based anti-cheating mechanisms.
- Integrated Gemini API to automatically generate examination questions, simplifying the instructor workflow.
- **Tech Stack:** Next.js, Express.js, PostgreSQL, Prisma ORM, Socket.IO, Gemini API

Smart ID Card Detection - CA-YOLOv8 | github.com/Nithwin/SICD | sicd.vercel.app

- Designed a customized YOLOv8 architecture for accurate real-time student ID card detection using CBAM attention and small-object detection enhancements.
- Built an end-to-end monitoring pipeline integrating InsightFace, OpenCV, Flask, and SQLite.
- Published the research as "CA-YOLOv8" in [IEEE Xplore \(2025\)](#).
- **Tech Stack:** Python, Custom YOLOv8, InsightFace, OpenCV, Flask, SQLite

Terminal TCP/IP Chat Network | github.com/Nithwin/Cpp_Projects

- Developed a real-time terminal-based chat application using TCP/IP socket programming.
- Implemented a client-server architecture supporting reliable bidirectional communication over raw sockets.
- **Tech Stack:** C++, Linux, Socket Programming, TCP/IP

ACHIEVEMENTS

- **Competitive Programming:** Solved 770+ Data Structures and Algorithms problems on LeetCode.
- **Research Publication:** Published CA-YOLOv8 in IEEE Xplore (2025).
- **Innovation Excellence (2× First Prize):** Won university innovation competitions for AI Classroom Behavior Monitoring and CA-YOLOv8.
- **Open Source:** Creator and maintainer of WindMist, an open-source AI coding assistant written in Go.
- **Certification:** Google Cloud Technical Series – AI in Action (2026)